

Machine learning optimization techniques include gradient descent, Adam optimizer, and stochastic methods that improve convergence rates significantly. These methods are fundamental to training deep neural networks effectively.

Neural networks require careful tuning of hyperparameters such as learning rate, batch size, and number of epochs. The choice of optimizer can dramatically impact training speed and final model performance.

Convolutional neural networks are particularly effective for image recognition tasks. They use filters to detect features at different spatial scales, building up from edges to complex patterns.

Recurrent neural networks and transformers are used for sequential data processing. Transformers have largely replaced RNNs due to their ability to parallelize computation and capture long-range dependencies.

Transfer learning allows practitioners to leverage pre-trained models for new tasks with limited data. Fine-tuning a large model on a small dataset often outperforms training from scratch.